Design and Specification in Upper Limb Prosthetics: A Research Report

Summary

The development of upper limb prosthetics is a complex field marked by a persistent gap between technological capabilities and the functional, ergonomic, and psychosocial needs of users. Despite significant advancements in mechatronics, control systems, and materials, high rates of device abandonment persist, frequently cited as between 20% and 56%. The primary drivers of rejection are discomfort (especially at the socket interface), excessive weight, limited functionality, and non-intuitive control. The literature defines the core design problem as a multi-objective challenge: to create devices that not only replicate the biomechanical function of the human hand but are also lightweight, comfortable, affordable, and acceptable to the user in daily life. A paradigm shift is underway, moving from purely technical, function-driven design towards user-centered, participatory, and co-creation methodologies that prioritize the lived experience of individuals with limb differences. Emerging digital technologies, particularly 3D scanning, parametric CAD, and additive manufacturing, are central to this shift, enabling the rapid prototyping and production of personalized, low-cost solutions that can be tailored to individual anatomy and preferences. However, significant gaps remain, including the lack of standardized evaluation frameworks, insufficient long-term, real-world user studies, and challenges in translating advanced research into clinically viable and robust systems.

Key Findings

A synthesis of the research reveals several critical findings that characterize the current landscape of upper limb prosthetic design and specification.

- **Persistent User-Technology Gap**: A significant disconnect exists between the features of available prosthetic devices and the actual needs and priorities of users. High abandonment rates are a direct consequence, primarily driven by failures in comfort, functionality, weight, and intuitive control. Technological sophistication does not consistently translate to improved user satisfaction or adoption.
- **Centrality of User-Centered and Ergonomic Requirements**: Across multiple studies and stakeholder groups (users, clinicians, relatives), comfort is consistently ranked as the most critical requirement, often followed by low weight and durability.
- **Conflicting Design Objectives**: A core challenge in prosthetic design is balancing conflicting requirements. The trade-off between functionality (e.g., degrees of freedom, grip force) and wearability (e.g., weight, size, power consumption, control complexity) is a recurring theme.
- **Rise of Digital Fabrication and Personalization**: Additive manufacturing (3D printing) and associated digital workflows (3D scanning, parametric CAD) are identified as transformative technologies. They enable the creation of low-cost, lightweight, and highly customized devices, facilitating rapid prototyping and patient-specific solutions that can improve fit, comfort, and aesthetics.
- **Lack of Standardized Evaluation**: The field lacks common evaluation criteria and standardized protocols for assessing prosthetic performance and user satisfaction in real-world contexts. This heterogeneity makes it difficult to compare devices, validate new designs, and generate the robust evidence needed to inform clinical practice and policy.
- **Control and Sensory Feedback Deficits**: A major limitation of current systems is the non-intuitive nature of control, often relying on limited EMG signals to operate multi-degree-of-freedom hands. Concurrently, the lack of meaningful sensory feedback limits dexterous control and embodiment.

Definitions

The literature defines the design problem for upper limb prostheses not as a single objective, but as a complex, multi-objective optimization challenge centered on restoring capability while respecting the user's lived experience. This problem is framed across several interconnected domains:

- **Biomechanical and Functional Replication**: At its core, the design problem involves creating an artificial limb that can replicate the lost functions of a biological hand and arm. This is defined by attempts to match anatomical kinematics and kinetics, including degrees of freedom (DOF), range of motion (ROM), grip patterns, force, and speed. The goal is to enable the user to perform essential Activities of Daily Living (ADLs) independently.
- **User-Centred Problem Solving**: A significant portion of the literature reframes the design problem away from pure technical replication towards meeting user-centric needs to reduce high rates of device abandonment. In this view, the problem is defined by the failure of existing devices to be comfortable, lightweight, intuitive, reliable, and aesthetically acceptable. Therefore, the design task becomes one of identifying and prioritizing user needs and translating them into design specifications that enhance satisfaction and quality of life.
- **Multi-Objective Optimization and Trade-offs**: Many sources define the problem as finding an optimal trade-off between conflicting requirements. A primary conflict exists between functionality (e.g., more actuators, higher forces) and wearability (e.g., lower weight, smaller size, lower power consumption). The design process is thus an exercise in navigating these trade-offs to arrive at a solution that is practical for real-world use.
- **Human-Machine Interfacing**: A critical aspect of the design problem is defined as restoring the bidirectional communication loop between the user's nervous system and the artificial limb. This includes both an efferent pathway (decoding user intent from neural or muscular signals for intuitive control) and an afferent pathway (providing sensory feedback to the user). The lack of sufficiently powerful and reliable interfaces is considered a major limiting factor in the field.

Requirement Types

The design and specification of upper limb prostheses involve a wide array of interconnected requirements spanning multiple domains.

Functional Requirements

These define what the prosthesis must do to restore capability. Key functional requirements include:

- **Grasping and Dexterity**: The ability to perform a variety of grip patterns (e.g., power, precision, pinch, lateral) with sufficient grip force and stability.
- **Degrees of Freedom (DOF) and Range of Motion (ROM)**: The number of independent joints and their angular travel, which should ideally mimic the natural limb to enable complex tasks and reduce compensatory body movements.
- **Speed and Control**: Actuation speed and control precision are critical for natural and effective interaction. This includes proportional control of both velocity and force.
- **Task Performance**: The ultimate functional goal is the ability to perform Activities of Daily Living (ADLs) such as eating, dressing, and handling common objects.

Ergonomic Requirements

Ergonomic requirements are crucial for user acceptance and are the most frequently cited reasons for device abandonment.

- **Comfort**: This is consistently ranked as the highest priority for users.
- **Weight**: Low weight is a primary user demand to reduce fatigue and improve wearability10,61,71,103,113,212,215,353,412,425,450,
- **Wearability and Usability**: This includes ease of donning and doffing, minimal noise during operation, and overall ease of use.

Technical Requirements

These specify the engineering components and systems needed to meet functional and ergonomic goals.

- **Actuators and Transmissions**: Selection of motors (e.g., DC, brushless), drive mechanisms (e.g., tendon-driven, linkages), and transmissions that can provide the necessary force and speed within size and weight constraints.
- **Sensors and Control Interfaces**: This includes myoelectric (EMG) sensors, force/pressure sensors, and sometimes more advanced interfaces (e.g., EEG, neural interfaces) to capture user intent. The control system must process these signals to command the actuators, often using pattern recognition or machine learning.
- **Power Source**: The battery must provide sufficient autonomy for a full day of use while being compact and lightweight.
- **Materials**: Materials must be lightweight, strong, durable, and biocompatible.

Manufacturing Requirements

These relate to the fabrication, assembly, and cost-effectiveness of the device.

- **Design for Manufacturing (DFM)**: Designs should be simple, scalable, and easy to assemble and repair.
- **3D Printing / Additive Manufacturing**: Increasingly, designs are optimized for fabrication using technologies like Fused Deposition Modeling (FDM) or Selective Laser Sintering (SLS), which allows for complex geometries and customization.
- **Modularity and Cost**: Modular designs can reduce costs and simplify repairs or upgrades. Overall device cost is a major barrier to access, particularly in developing countries.

User-Centred Requirements

These requirements go beyond basic function to encompass the user's social and psychological needs.

- **Aesthetics and Cosmesis**: The visual appearance of the prosthesis is highly important to many users, influencing social acceptance and body image.
- **Personalization**: Users desire devices that are tailored to their specific needs, activities (e.g., sports), and preferences.
- **Psychosocial Factors**: The design should promote confidence, embodiment (feeling that the prosthesis is part of the body), and social participation, addressing factors that lead to rejection.

Regulatory and Standards Requirements

Prosthetic devices, as medical devices, must adhere to regulatory standards to ensure safety and quality. Key standards include:

- **ISO 13485**: Specifies requirements for a quality management system for medical device manufacturers, covering design, development, production, and service.
- **ISO 22523**: Outlines requirements and test methods for external limb prostheses, covering strength, materials, risk management, clinical evaluation, and electrical safety.

- **IEC 60601**: A series of technical standards for the safety and essential performance of medical electrical equipment.

Dominant Approaches and Theoretical Frameworks

The design and specification of upper limb prostheses are guided by several dominant methodologies and theoretical frameworks that have evolved from traditional engineering processes to more integrated, user-focused approaches.

A foundational approach is the **systematic engineering design process**, often based on models like those proposed by Pahl and Beitz, which involves structured phases of planning, conceptual design, embodiment design, and detail design. This is complemented by a **requirements-driven design procedure**, where a comprehensive list of user needs and technical specifications is established at the outset to guide development.

More recently, there has been a significant shift towards **User-Centered Design (UCD)** and **Human-Centered Design (HCD)**. UCD is often implemented through an iterative process involving user feedback at multiple stages. This is closely related to **Participatory Design** and **Co-creation**, where end-users, clinicians, and other stakeholders become active partners in the design process, not just subjects of study.

To translate user needs into technical specifications, some methodologies employ structured tools. **Quality Function Deployment (QFD)**, and its central tool, the **House of Quality**, is used to map customer requirements to engineering characteristics, ensuring that technical design decisions are directly linked to user priorities. The **Structure-Behavior-Function (SBF) ontology** is another theoretical framework used to model the relationship between a product's intended function, its behavior, and its physical structure, aiding in the conceptual design phase.

Common Technical Strategies and Modelling Methods

To translate design requirements into functional prototypes, a variety of technical strategies and modelling methods are employed, often combining digital tools with physical testing.

Digital Modelling and Simulation

- **Computer-Aided Design (CAD)**: CAD software is fundamental to the design process. Tools like SolidWorks, Autodesk Fusion, and CATIA are widely used for creating detailed 3D models of prosthetic components and assemblies. These models serve as the basis for simulation, manufacturing, and visualization.
- **Finite Element Analysis (FEA)**: FEA is used to simulate how a design will behave under real-world forces, allowing engineers to analyze stress, strain, and deformation. This is crucial for optimizing strength-to-weight ratios and ensuring structural integrity, especially for load-bearing components. **Topology optimization** is an advanced simulation technique used to algorithmically remove material from a part while maintaining its structural performance, leading to highly efficient, lightweight designs.
- **Kinematic and Kinetostatic Modelling**: These methods are used to analyze the motion and forces within the prosthetic system. Techniques like Denavit-Hartenberg (D-H) parameters and multibody dynamics simulations (e.g., in OpenSim or ADAMS) help define joint ranges, calculate required actuator torques, and ensure the mechanism can perform desired tasks.

Personalization and Fabrication

- **3D Scanning and Reverse Engineering**: To create patient-specific devices, 3D scanners are used to capture the geometry of the user's residual limb. This digital data provides the foundation for designing a custom-fitted socket, which is a critical element for comfort and function.

- **Parametric Modelling**: This powerful strategy involves creating designs where key dimensions are controlled by parameters rather than fixed values. Software like Grasshopper is used to build algorithms that can automatically generate a customized design based on a user's specific measurements. This approach is central to mass customization and automated design pipelines.
- **Additive Manufacturing (3D Printing)**: 3D printing is the primary manufacturing method for producing customized, low-cost, and lightweight prosthetic components. Technologies like Fused Deposition Modeling (FDM) are common for low-cost prototypes, while Selective Laser Sintering (SLS) and Multi Jet Fusion (MJF) are used for more durable, higher-quality parts.

Identified Challenges and Limitations

Despite progress, the development of upper limb prosthetics faces significant and persistent challenges that limit device effectiveness and user acceptance.

A primary challenge is navigating the inherent **trade-offs between conflicting requirements**. The most frequently cited conflict is between functionality and wearability, where adding more degrees of freedom, actuators, and sensors to increase dexterity inevitably increases the device's weight, size, power consumption, and control complexity. This often leads to devices that are either functionally limited or too heavy and cumbersome for daily use.

Control complexity remains a major barrier. Most advanced prostheses rely on surface electromyography (sEMG), but there are a limited number of independent muscle sites available for control, which is insufficient to intuitively command the many degrees of freedom of a modern prosthetic hand. This mismatch requires users to engage in cognitively demanding switching between grip patterns and results in control that is often perceived as awkward and unnatural. Furthermore, sEMG signals are notoriously unreliable and can be affected by factors like sweat, electrode shift, and muscle fatigue.

The **prosthetic socket**, which is the interface between the device and the user's residual limb, is a critical source of problems. Poor socket fit is a leading cause of discomfort, pain, and skin issues, and is a primary reason for device abandonment. Designing and fabricating a well-fitting socket is a time-consuming and skill-intensive process that is difficult to standardize.

Finally, systemic issues such as **high cost**, limited access to technology, and a significant gap between laboratory research and commercially available products present major limitations. Many advanced control strategies and sensory feedback systems developed in academic settings are not robust or cost-effective enough for clinical translation, leaving a persistent gap between research potential and real-world benefit for amputees348,765,783,859.

Gaps in Current Research

The literature reveals several critical gaps in the research and development of upper limb prosthetics that hinder progress and contribute to the persistent user-technology divide.

A primary gap is the **lack of standardized frameworks and evaluation metrics**. There are no universally accepted protocols for assessing the functional performance of prosthetic hands in a way that is comparable across different studies and devices. Existing outcome measures are often not specific to the upper limb population, fail to provide a holistic view of success, and rarely integrate patient-reported feedback in a structured manner. This makes it difficult to objectively validate new designs and generate the high-quality evidence needed to guide clinical practice and policy.

There is a significant deficiency in **long-term, real-world user studies**. Most research is conducted in controlled laboratory settings over short periods, which often fails to capture the complexities of daily device use, including durability, reliability, and the impact of environmental factors.

Insufficient user involvement in the design process remains a critical issue. While user-centered design is increasingly advocated, its implementation is often superficial. A deeper integration of co-creation methodologies, where users are partners throughout the entire research and development lifecycle, is necessary to ensure that technological advancements are aligned with genuine user needs and priorities380,735,744,814.

Furthermore, specific populations, such as **bilateral amputees**, are often under-researched, with their unique needs and challenges not adequately addressed in design considerations. Finally, there is a recognized **translational gap**, where many innovative technologies developed in academic labs, particularly in areas like advanced control and sensory feedback, fail to transition into robust, affordable, and clinically viable products95,348,405,480,765,859.

Implications for Parametric and Personalised Prosthetic Development

The widespread adoption of digital technologies, particularly 3D scanning, parametric modeling, and additive manufacturing, has profound implications for the development of personalized upper limb prosthetics. These tools are shifting the paradigm from standardized or manually customized devices to automated, patient-specific solutions.

3D scanning allows for the rapid and accurate capture of a user's residual limb geometry. This digital data forms the foundation of a personalized design workflow, enabling the creation of prosthetic sockets with a precise, comfortable fit that would be difficult and time-consuming to achieve with traditional casting methods801,1355,1357,1369.

Parametric Computer-Aided Design (CAD) is a key enabler of mass customization. By creating a base model where critical dimensions are driven by parameters (e.g., limb length, circumference, socket shape), designers can automate the process of generating a unique device for each user. Frameworks using software like Grasshopper can even be hosted online, allowing clinicians or users to input measurements and generate a ready-to-print 3D model, dramatically reducing costs and design time or adapted to different amputation levels1258,1263.

The concept of a **digital twin** or a **statistical shape model (SSM)** represents a more advanced application. These models can predict internal anatomical structures from external surface scans, allowing for the creation of FEA models to simulate the limb-socket interface and optimize pressure distribution before fabrication1274,1324,1325,1333,1339. This predictive capability promises to move prosthetic fitting from a reactive, iterative process to a more proactive, evidence-based one.

Finally, **additive manufacturing (3D printing)** makes it possible to physically realize these complex, personalized designs in a cost-effective manner. It allows for the fabrication of integrated mechanisms, complex internal lattice structures for cushioning and ventilation, and aesthetically unique designs, all tailored to a single user952,953,957.

Recommendations

Based on the synthesized findings, several evidence-based recommendations can be made to researchers and practitioners to advance the design and specification of upper limb prosthetics.

For Researchers:

- **Adopt and Standardize User-Centered Methodologies**: Prioritize the integration of robust user-centered design, participatory design, and co-creation frameworks into research projects380,744,814. There is a critical need to develop and validate standardized, mixed-methods approaches for evaluating prosthetic performance

that combine objective functional metrics with patient-reported outcomes on comfort, usability, and quality of life35,90,346,486,638,648.
- **Conduct Longitudinal, In-Home Studies**: Shift the focus of evaluation from short-term laboratory assessments to long-term studies in users' natural environments244,729,739,747,842. This is essential for understanding real-world device usage, durability, and the factors that contribute to sustained adoption or abandonment.
- **Address Key Technical Gaps**: Focus research efforts on critical unsolved challenges. This includes developing more intuitive and robust human-machine interfaces that overcome the limitations of sEMG, creating effective and non-invasive sensory feedback systems, and improving the energy density and longevity of power sources862,863,864.
- **Promote Open Science and Data Sharing**: The lack of large, diverse datasets hinders the development of generalizable models and benchmarks494,1318,1321,1322. Creating shared databases of anatomical scans, biomechanical data, and standardized test results would accelerate progress in the field445,1323.

For Practitioners and Engineers:

- **Embrace Digital Design and Manufacturing Workflows**: Leverage 3D scanning, parametric CAD, and additive manufacturing to create personalized, better-fitting, and more affordable prosthetic devices770,796,947,961. Investing in training for these digital tools is crucial for modern prosthetic practice806.
- **Prioritize Fundamental User Needs**: While advanced functionality is an important goal, design efforts must first address the primary reasons for device abandonment: comfort, weight, and reliability10,61,166,308,447,450. Iterative prototyping with frequent user feedback is key to optimizing these factors.
- **Design for Modularity and Adaptability**: Develop modular prosthetic systems that can be easily customized, upgraded, or repaired151,232,481,501,611. This approach allows devices to be adapted to an individual's changing needs and facilitates more sustainable and cost-effective service delivery models.
- **Foster Interdisciplinary Collaboration**: Effective prosthetic design requires a transdisciplinary approach that integrates expertise from engineering, clinical rehabilitation, industrial design, and computer science249,449,751. Establishing clear communication and shared goals among these diverse stakeholders is essential for translating user needs into successful products.

By focusing on these areas, the field can better bridge the gap between technological possibility and the lived reality of users, leading to the development of upper limb prostheses that are not only more functional but also more readily accepted and integrated into daily life.